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DCCF: Dynamic co-optimizing client selection and gradient compression for efficient-robust FL under non-IID data

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ABSTRACT

Federated Learning (FL) enables decentralized collaborative training method without compromising privacy of participants. However, in Non-Independent and Identically Distributed (Non-IID) data scenarios, it often encounters issues including delayed convergence, unstable performance, and high communication costs. Existing approaches face the following limitations: (1) client selection mechanisms often rely on predetermined trade-off strategies that lack adaptability to dynamic shifts in client contributions during training; and (2) many mainstream frameworks either adopt a sequential "select-then-compress" paradigm, or rely on system performance metrics (e.g., computation latency, channel gain). Such designs struggle to client selection that is inconsistent with the gradient compression objective, making it difficult to balance performance and efficiency under data heterogeneity. This paper proposes a Discrepancy-Metric based Client Selection and Compression Framework (DCCF), its core innovation resides in dynamically coordinating client selection and personalized compression through a unified discrepancy metric, achieving dynamic synergistic optimization of both. Specifically, the framework first synchronously assesses the consistency between the local data distribution of each client and the global model via this discrepancy metric, then adaptively selects participating clients based on the evaluation results. This approach balances model consistency and data diversity, thereby suppressing the drift of local models. Furthermore, personalized compression ratios are assigned to each selected client using the discrepancy metric: clients with low discrepancy values are allocated high compression ratios to conserve communication bandwidth, while those with high discrepancy are assigned low compression ratios to preserve crucial gradient information. Additionally, the framework integrates a residual accumulation mechanism to alleviate compression-induced errors and enhance the stability of the training process. More importantly, instead of naively combining client selection and gradient compression, we highlight that their decisions are tightly coupled. Both theoretical analyses and experimental validations demonstrate the necessity of such joint optimization. Experimental results indicate that DCCF outperforms existing mainstream methods in terms of model accuracy, convergence speed, and communication efficiency. Particularly under the extremely Non-IID scenario of the CIFAR-10 dataset, DCCF not only reduces the communication overhead drastically to 7.9% of the baseline method but also accelerates the model convergence rate by 56%.

1. Introduction

The swift progress of Internet of Things (IoT) in recent years has facilitated the extensive adoption of machine learning across multiple domains, including smart home (Taiwo et al., 2022) and drone technology (Chen et al., 2024a). Each edge device generates substantial amounts of on-device data, providing rich resources for machine learning (Hua et al., 2023; Kong et al., 2022). As a prominent distributed paradigm framework, Federated learning has advanced to allow multiple end devices to jointly refine models locally without the need for raw data exchange, thereby preserving privacy.

Though advantageous, Federated Learning (FL) encounters the challenge of Non-Independent and Identically Distributed(Non-IID) data across participating devices (Li et al., 2022). In practice, edge de-

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vices often collect data with divergent feature distributions, leading to substantial discrepancies in local data statistics. This statistical heterogeneity induces deviations of client-side local updates from the global optimization objective, thereby triggering local model drift. Such a phenomenon significantly leads to the model's convergence rate and final prediction accuracy while exacerbating the instability of the training process (Li et al., 2025; Liu et al., 2021; Ma et al., 2022). Specifically, conflicting updates among clients often require additional communication rounds to ensure the effective integration of model parameters. This added coordination not only increases communication overhead but also compromises global efficiency, particularly in resource-constrained environments. To address these challenges, significant research has been conducted to optimize the effectiveness and robustness of FL in Non-IID data scenarios. To mitigate the adverse effects of Non-IID data, several sampling methods have explored distinct criteria for evaluating client importance, including local loss (Sultana et al., 2022), test accuracy (Zhu et al., 2024), model updates (Balakrishnan et al., 2022; Zhang et al., 2021), and client relevance (Tang et al., 2022; Zhao et al., 2026). In each training round, clients holding "important" data are selected with a higher probability. However, these methods often rely on fixed selection criteria or static trade-off strategies, which renders them unable to adapt to the dynamic changes in client contributions throughout the training process. Furthermore, these approaches typically require clients to upload complete model updates, introducing substantial communication overhead in bandwidth-constrained edge networks.To alleviate this burden, techniques such as sparsification (Cui et al., 2022; Xu et al., 2021) or quantization (Qu et al., 2024; Xu et al., 2023) have been employed to cut down the communication load.Nevertheless, most of these compression schemes assign fixed or uniform compression ratios to all clients, overlooking the inherent heterogeneity in data distributions across clients. Recent studies have attempted to jointly optimize client selection and gradient compression. Some frameworks (e.g., FedPAQ Reisizadeh et al., 2020, RFCSC Pan et al., 2025) adhere to a sequential "select-then-compress" paradigm. This paradigm may lead to severe loss of critical gradient information from selected clients during the subsequent fixed compression process, thereby significantly diminishing the effectiveness of the selection strategy itself in mitigating statistical heterogeneity. Additionally, Chen et al. (2024b) regard FL as a resource scheduling problem, where clients are treated as resource nodes (with speed and bandwidth), data as workload, and the goal is to complete the training task efficiently. The utility function of this approach relies heavily on system performance metrics (e.g., computation latency, channel gain), which gives rise to a fundamental contradiction: under highly Non-IID scenarios, clients with poor channel conditions may nonetheless possess valuable data diversity. Conversely, clients with excellent channels may provide updates that are less informative or even detrimental to the global model. Assigning low compression ratios to such clients may instead exacerbate model drift.

In summary, existing research has failed to comprehensively address the aforementioned challenges, thereby hindering the realization of efficient Federated Learning.Consequently, designing a unified framework that tightly couples client selection with adaptive compression remains an open and critical research question.

However, the joint optimization of client selection and compression ratios poses considerable challenges, primarily for the following two reasons: First, the inherent relationship between client selection, gradient compression, and model convergence remains incompletely elucidated. Second, the compression ratio exerts two mutually conflicting impacts on the training process, rendering it challenging to determine an optimal compression ratio that balances resource overhead and model prediction accuracy.

To address these challenges, this paper proposes the Discrepancy-Metric based Client Selection and Compression Framework(DCCF). Distinct from existing approaches, DCCF focuses on capturing the dynamic interplay between client selection and gradient compression, with the goal of minimizing communication overhead while improving the overall performance of the global model. Specifically, DCCF integrates a discrepancy metric into the client selection process, which synchronously evaluates the discrepancy between individual client data distributions and the global model. This metric directly drives two tightly coupled decisions: on one hand, client selection prioritizes clients with the highest consistency (with the global model) to ensure update quality; on the other hand, it incorporates random client selection to enhance data diversity. This design achieves an adaptive trade-off between model consistency and data diversity, thereby effectively suppressing local model drift induced by statistical heterogeneity. Furthermore, leveraging the same discrepancy metric, a personalized compression ratio is assigned to each selected client: clients with smaller discrepancy (i.e., their local data is more generalizable) are allocated higher compression ratios to conserve communication bandwidth, whereas those with larger discrepancy (i.e., their local data contains more unique information) receive lower compression ratios to preserve critical gradient information. This strategy ensures that communication overhead is well-controlled while safeguarding the integrity of the most informative gradients. Moreover, to compensate for the information loss caused by gradient compression, DCCF introduces a residual accumulation compensation mechanism. This mechanism accumulates the untransmitted gradient information into subsequent training iterations, which in turn ensures long-term training stability and the final prediction accuracy of the global model. The main contributions of this paper are as follows:

- We propose a unified joint optimization framework based on discrepancy metric, breaking away from the traditional sequential paradigm of "select first, then compress." This framework enables simultaneous and collaborative decision-making between client-side selection and gradient compression, effectively addressing model drift and communication efficiency issues posed by data heterogeneity.
- We design a personalized resource allocation mechanism based on discrepancy metrics. By utilizing the discrepancy value between each client-side trained models and the central aggregated model as an indicator of data heterogeneity, we achieve adaptive decision-making for client selection and compression ratio allocation. This approach strikes a better balance between global model performance and communication efficiency.
- We introduce a residual accumulation mechanism to mitigate compression errors and validate the framework's effectiveness through both theoretical analysis and experimental verification. By accumulating and compensating for untransmitted gradient information, we enhance training stability and model accuracy. Experiments across multiple benchmark datasets demonstrate that DCCF significantly outperforms existing methods in terms of model accuracy, convergence speed, and communication efficiency.

The subsequent sections of the paper are organized in the following manner: Section 2 offers a comprehensive review of prior research.The problem formulation is presented in Section 3. Section 4 elaborates on the proposed algorithm design. Performance evaluation and experimental results are discussed in Section 5. Finally, Section 6 concludes the paper with a summary and future research directions.

2. Related work

The presence of Non-IID data presents a major obstacle in federated learning, particularly in resource-constrained environments propelled by the Internet of Things. Given the heterogeneity of edge terminals, the local data distributions across clients exhibit marked statistical disparity, resulting from different usage patterns and data generation methods. This heterogeneity gives rise to deviations of clients' local models from the global model.

Early FedAvg (McMahan et al., 2017) adopted random uniform sampling,which completely ignored differences in data distributions across clients and thus failed to adapt to training requirements in Non-IID scenarios. Zhao et al. (2018) proposed distributing public data to clients

to make the clients' data distributions closer to being independent and identically distributed. Subsequently, client selection methods based on contribution or data scale were proposed. On the data side, Pandey et al. (2022) employed a contribution-aware device selection scheme that guides selection by estimating the marginal contribution of devices to model performance. However, these approaches typically rely on static evaluation criteria, rendering them unable to respond to dynamic changes in client contributions throughout the training process.In recent years, model-based selection strategies have garnered attention, such as those using local loss (Sultana et al., 2022) or test accuracy (Zhu et al., 2024) as evaluation metrics. Nevertheless, this type of selection mechanism overlooks the diversity of data distributions, potentially leading to the exclusion of important but data-scarce clients and thereby compromising model quality. More flexible selection mechanisms, such as FedAEB recently proposed by Zheng et al. (2024), leverage reinforcement learning to dynamically optimize client selection, thereby significantly improving convergence efficiency. FedAgent (Sun et al., 2025) employs a dueling network and double DQN to intelligently select participating clients, and further couples client selection with knowledge distillation to alleviate client drift. While this approach demonstrates improved convergence and accuracy on Non-IID data, it does not address the communication efficiency challenge from the perspective of gradient compression. While these methods can dynamically reflect client contributions to the global model, they typically require clients to upload complete model updates, which imposes substantial communication overhead in bandwidth-constrained scenarios.

In federated learning, gradient compression has become a focal point for addressing communication bottlenecks. Early studies focused on reducing communication through gradient sparsification. For instance, Lin et al. (2017) developed Deep Gradient Compression (DGC), utilizing momentum correction and local gradient clipping to attain high sparsity. However, this generally results in the loss of gradient information when sparsity is too high, thereby affecting model convergence. In contrast, Sattler et al. (2019) proposed Sparse Ternary Compression , which achieves ultra-low bit-width communication while demonstrating robust convergence even under highly Non-IID data distributions. However, most compression schemes assign fixed or uniform compression ratios to all clients, overlooking the inherent heterogeneity in data distributions across clients. While subsequent studies have attempted to introduce dynamism into compression strategies, for example, Han et al. (2020) explored adaptive sparsity by considering the characteristics of federated learning tasks. DeepReduce (Xu et al., 2021) decomposes sparse tensors into values and indices, enabling independent and combined compression of these components. Hönig et al. (2022) proposed a dynamic adaptive quantization mechanism that assigns differentiated quantization bits to different model layers by analyzing historical gradient distributions. Yet these approaches still fail to effectively link compression strategies to the data importance of individual clients.Notably, recent works have attempted to jointly optimize client selection and gradient compression, such as Chen et al. (2024b). These approaches primarily target the optimization of communication fairness and transmission latency. Their joint selection-compression decisions rely on external communication conditions (e.g., channel gain) rather than the intrinsic differences between client data and the global model. Consequently, such methods fundamentally fail to address model drift induced by Non-IID data distributions. Reisizadeh et al. (2020) proposed FedPAQ, which reduces communication overhead through quantization, periodic averaging, and partial device participation. However, FedPAQ adopts a sequential processing paradigm: client selection and quantization operations are performed independently, and the quantization strategy remains unlinked to the data importance of selected clients. This decoupled optimization approach leads to potential information loss during subsequent quantization for clients deemed important by the selection mechanism, thereby undermining the effectiveness of the selection process itself. Similarly, Chen et al. (2024b) establishes a unified system utility function to place client selection and gradient compression under a joint optimization paradigm; however, its optimization objective relies heavily on system efficiency metrics centered on communication quality. Under this decision-making mechanism, updates with large directional deviations are systematically selected and transmitted in full. During iterative aggregation, these updates with systematic deviations continuously disrupt the global optimization trajectory–they not only fail to effectively promote convergence but also exacerbate local model drift among clients, making it difficult to achieve synergistic improvements in convergence and communication efficiency under Non-IID environments. Building upon prior research, this paper proposes a discrepancy-based client selection and compression framework. Its core innovation resides in realizing synergistic dynamic optimization of client selection and gradient compression via discrepancy metrics, thereby addressing the key challenges of slow convergence, unstable model performance, and high communication costs arising from data heterogeneity.

3. Problem analysis

3.1. Model drift

In federated learning, the diversity in client data distribution stands as a pivotal challenge, contributing to the decline in global model performance. Assuming the existence of N clients, the local objective function of client is denoted as:

$$F_n(w) := \frac{\sum_{\xi \in D_n} f_n(w, \xi)}{|D_n|} \tag{1}$$

where the local objective $F_n(w)$ of client n computed over its collected data samples, is given by the empirical mean of $f_n(w, \xi)$, here, w is a model parameters and ξ is the local training data samples, $|D_n|$ is the size of the local dataset of client n. FL aims to minimize its loss objective, specifically, to solve for:

$$\min_{w} \left\{ F(w) := \sum_{n=1}^{N} \pi_n F_n(w) \right\} \tag{2}$$

where π_{n} is defined as the weight of client n and $\sum_{\mathrm{n}=1}^{N} \pi_n = 1$.

As shown in Fig. 1, in the IID setting, the client models are updated toward their respective local optima and the average model w_f^* approaches the ideal global model w^*. However, when data follows a Non-IID distribution, these local models are not only far apart from one another but also deviate significantly from the global optimum. Even though all clients start at the same point, each one moves towards its own optimum. The difference between the server-side optimum obtained by classical aggregation strategies such as federated averaging and the true optimum produces a large distance.

3.2. Compression error

In the realm of federated learning, top-k sparsification is extensively utilized as a gradient compression method to significantly decrease communication costs. Consider client n with its original gradient vector g_n. Top-k compression generates an approximated gradient $\hat{\mathrm{g}}_n$ by retaining the k elements with the largest absolute values from g_n and nullifying the rest, thus streamlining the transmission of the gradient matrix. However, this operation inherently sacrifices gradient precision, consequently decelerating convergence and compromising the federated model's aggregate performance. Determining the optimal sparsity level remains a formidable challenge, largely due to the distinct nature of federated learning tasks and variations among client models. A high compression ratio may limit precision loss but falls short in significantly enhancing communication efficiency. On the contrary, a low compression ratio risks discarding crucial historical gradient information, causing the model update trajectory to deviate from the true gradient direction. This misalignment substantially decelerates convergence and degrades the

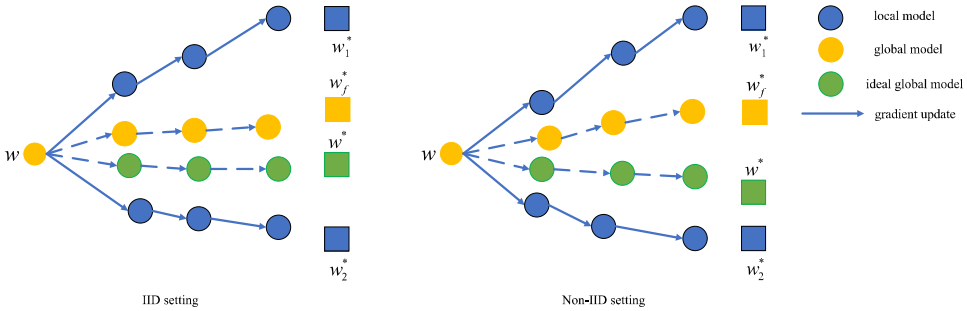


Fig. 1. A sketch for the impact of Non-IID data.

effectiveness of the collaborative model. Under heterogeneous data environments, the uneven distribution of gradients exacerbates the residual compensation bias, further impeding the model's convergence process.

4. Methodology design

4.1. Client selection

A core challenge in the proposed federated learning framework is to efficiently and continuously select clients with low data heterogeneity for training participation, thereby mitigating the bias introduced by highly non-IID clients. Simply selecting clients at random in each round is suboptimal, as it overlooks valuable historical information regarding client data distributions. Conversely, solely relying on a fixed set of "good" clients risks overfitting and lacks exploration for potentially better candidates. Therefore, the federated server optimizes the client selection process by maintaining three key client sets to enhance the performance of model training. These sets are the random set N_t, the candidate client set C_t, and the final client set S. To elaborate, N_t s the set of clients randomly selected in round t to explore new directions for model updates; C_t is the candidate set containing clients that may be finally selected; and S is a client subset selected to ultimately engage in model aggregation following the screening process.

Initially, the proportion of randomly selected clients is defined as $\mathcal{P}$, meaning the size of N_t is $\mathcal{P}K$, where K represents the total number of available clients. Meanwhile, the final selected set S contains at most ∂K, where ∂ is the selection factor. An auxiliary dataset participates in model training as an additional client (denoted as a_0), aiding in distinguishing differences among client models. In each training round, the federated server operates through the following steps:

The server performs model aggregation exclusively using clients from in N_t and S when the size of S falls below the required threshold. The aggregation formula is:

$$w^t = \sum_{c_n \in N_t \cup S} \frac{|D_n|}{|D|} w_n^t \tag{3}$$

where w_n^t denotes the model parameters of client n at t round, and $|D|$ represents the aggregate size of all participating clients' local datasets..

If S contains a sufficient number of clients, the server will add the auxiliary model to the global model:

$$w^t = \frac{|D_0|}{|D|} w_0^t + \sum_{c_n \in N_t \cup S}^{N} \frac{|D_n|}{|D|} w_n^t \tag{4}$$

The server evaluates discrepancy variations between the refreshed auxiliary model and each client model. For instance, the normalized L2-norm of model weights between client n and auxiliary client a_0 is as follows:

$$\mathrm{D}_n^L = \frac{\left\| w_n^t - w_0^t \right\|}{\left\| w_0^t \right\|} \tag{5}$$

Specifically, we employ this discrepancy metric as a proxy indicator to quantify the distributional difference between each client's local data and the global model. Based on the computed discrepancy metrics, the server ranks clients within the candidate set $N_t = \left\{ c'_{n_1}, \quad c'_{n_2}, \quad \ldots \right\}$, where $D^t_{n'_1} \le D^t_{n'_2} \le \ldots$ Here, c'_{n_i} denotes the participating clients in the candidate set, so as to prioritize those with smaller discrepancies from the auxiliary model. Specifically, if the size of the final client set S has not reached the predefined maximum ∂K, the server selects clients from the candidate set C_t whose discrepancy differences are lower than those of the top $\psi \times 100\%$ clients in N_t. Subsequently, these selected clients are then added to the final set S.

4.2. Adaptive model compression

To further enhance communication efficiency and model performance, we introduce an adaptive compression strategy. The core of this strategy is a personalized compression ratio p_n^t and a residual vector R_n^t, which are defined for each client n at communication round t. Here p_n^t is a scalar denoting the fraction of gradient parameters to be retained. The residual R_n^t is a vector that accumulates the compression error over time, its dimension is identical to the model parameter vector w. In iteration t, the server prioritizes the top $\psi \times 100\%$ of clients from the random set N_t based on the L2 norm metric. Based on the difference metric, the server assigns a personalized compression ratio to each selected client. The compression rate for client i is determined by the following formula:

$$p_i = \begin{cases} p_{\max}, & \text{if } D_n^{\mathrm{L}} \le \tau_{\mathrm{low}} \\ p_{\min}, & \text{if } D_n^{\mathrm{L}} > \tau_{\mathrm{high}} \\ p_{\min} + (p_{\max} - p_{\min}) \cdot \frac{\tau_{\mathrm{high}} - D_n^{\mathrm{L}}}{\tau_{\mathrm{high}} - \tau_{\mathrm{low}}}, & \text{otherwise} \end{cases} \tag{6}$$

Here, τ_{high} is the maximum discrepancy value of the $\psi \times 100\%$ client before the random set N_t, and τ_{low} is the minimum discrepancy value of the candidate set C_t.

Following each round of local training, the client compresses the flattened parameter tensor X into a ternary-valued sparse tensor with elements in $\{-\mu, \quad 0, \quad \mu\}$. To mitigate compression-induced accuracy loss, we propose a residual error compensation mechanism. Specifically, during each compression step, the client first merges the residual R_n^{t-1} from the previous round with the current model parameters w_n^{t-1} to obtain the uncompressed local model Δw_n^{t-1} denoted as:

$$\Delta w_n^{t-1} = w_n^{t-1} + R_n^{t-1} \tag{7}$$

Subsequently, client n sparsifies the merged model parameters Δw_n^{t-1} to generate the compressed model $\Delta \widetilde{w}_n^t$ denoted as:

$$\Delta \widetilde{w}_n^t = top_k\left(\Delta w_n^{t-1}\right) \tag{8}$$

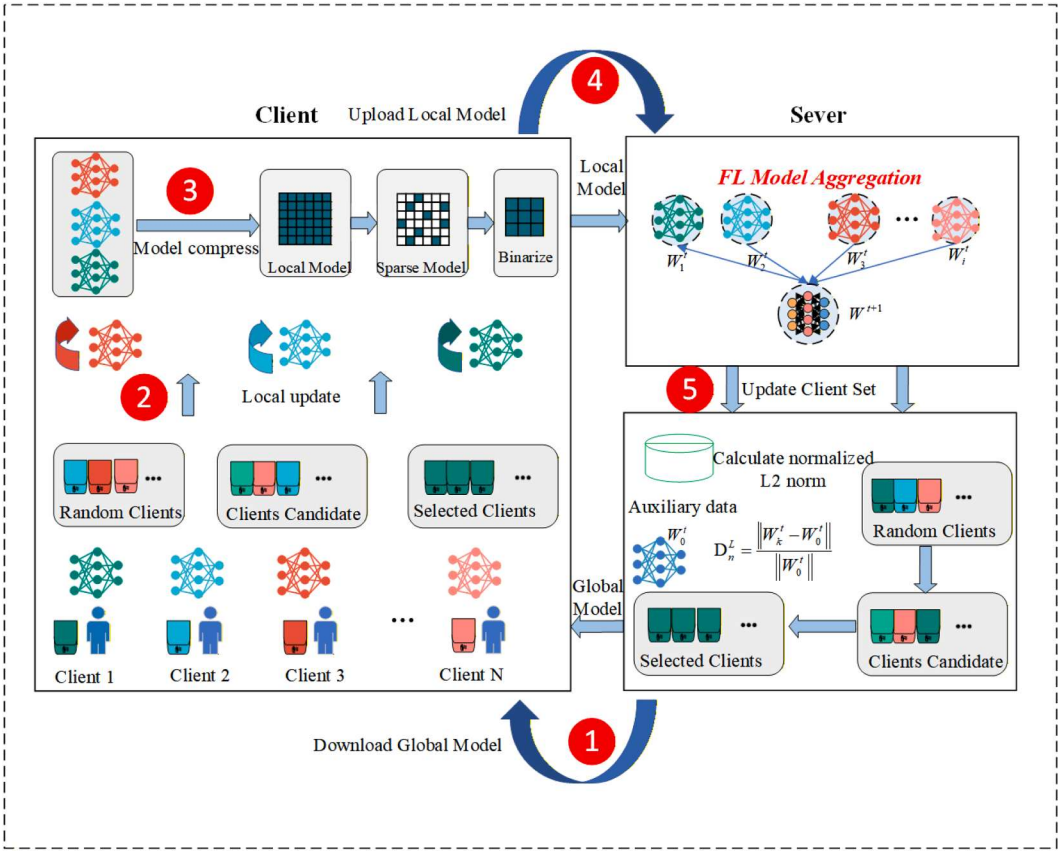


Fig. 2. Overview of DCCF.

The complete procedure is formally described in Algorithm 1. The uncompressed gradient components are accumulated in the residual term for subsequent training rounds:

$$R_n^t = R_n^{t-1} + \Delta w_n^{t-1} - \Delta \widetilde{w}_n^{t-1} \tag{9}$$

Algorithm 1 Adaptive model compression.

Require: flatten tensor X_i,compression ratios p_i
Ensure: : flatten ternary tensor$\widehat{X}_i \in \{-\mu, \quad 0, \quad \mu\}$
1: $k \leftarrow max(np_i,\ 1)$
2: $mask \leftarrow X_{\mathrm{i}} \geq topk(|\ X_{\mathrm{i}}\ |) \in \{0,1\}^{\mathrm{n}}$
3: $\widetilde{X}_{\mathrm{i}} \leftarrow \mathrm{mask} \odot X_{\mathrm{i}}$
4: $\mu \leftarrow \frac{1}{\mathrm{k}} \sum_{\mathrm{i}=1}^{\mathrm{n}} |\widetilde{X}_{\mathrm{i}}|$
5: $\widehat{X}_{\mathrm{i}} \leftarrow \mu \times \mathrm{sign}\left(\widetilde{X}_{\mathrm{i}}\right)$
6: Return $\widehat{X}_{\mathrm{i}}$

4.3. Framework for DCCF

The DCCF Framework fundamentally departs from the traditional sequential processing paradigm by tightly integrating client selection and gradient compression into a co-optimized module. Its core innovation lies in introducing a unified discrepancy metric, which simultaneously assesses the consistency between each client's local data distribution and the global model, while guiding both critical decisions: client selection and gradient compression. Specifically, during the client selection phase, this discrepancy metric is used to identify and retain the subset of clients that are both most consistent with and most diverse relative to the global model. In the compression phase, the same metric directly determines a personalized compression ratio for each selected client. This design ensures real-time alignment between client selection and compression strategies in every training iteration, effectively addressing the challenges posed by Non-IID data.

As illustrated in Fig. 2 and detailed in Algorithm 2, the framework initiates with a global model and an auxiliary model trained on a small, representative IID dataset which may originate from a public source residing on the server. This auxiliary model serves as a reference point for assessing the distributional alignment of each client. Each communication round of DCCF consists of the following orchestrated phases:

- Global Model Distribution. Upon initiating a new round, the server broadcasts the global model $\widetilde{w}^{t-1}$ to every participating client (line 2).
- Intelligent Client Selection. Prior to the local update, the server generates an initial set N_t through random selection and then evaluates the discrepancy between each client's local model and the auxiliary model. Then server applies our discrepancy-based to sift out the most representative subset C_t and S from the entire client population (lines 3-4).
- Client Local Update. Each client i in subset N_t , C_t and S , performs local training by minimizing the proximal term loss function based on the global model(line 6).
- Adaptive Model Compression with Residual Accumulation. Following local training, each client dynamically adjusts the compression ratio p_i based on discrepancy metrics to conduct sparsification on the original model update Δw_i^t, resulting in the compressed update $\Delta \widetilde{w}_i^t$. To further minimize the data volume, this compressed update is quantized prior to being uploaded to the server. The residual structure preserves the integrity of the gradient information, and any elements not transmitted are accumulated and will contribute to subsequent sparse computations(lines 7-10).

Algorithm 2 The design of discrepancy-metric based client selection and compression framework (DCCF).

Require: $t = 0$, $w^t, w_0^t \leftarrow w^t, S \leftarrow \{\}, C_t \leftarrow \{\}$
1: **for** each communication round $t = 1, 2, \ldots$ **do**
2: The Server sends the current global model w^t to all clients
3: $N_t \leftarrow$ Randomly select ρK clients
4: Update Client Set C_t, S
5: **for** client $c_n \in N_t \cup C_t \cup S$ **in parallel do**
6: $w_n^t \leftarrow$ ClientUpdateWithFedProx(c_n , w^{t-1}, μ)
7: $\Delta w_n^t \leftarrow$ by Eq. (7)
8: $\Delta \tilde{w}_n^t \leftarrow$Adaptive Model Compression
9: $R_n^t \leftarrow$ by Eq. (9)
10: **end for**
11: The server updates w_0^t using w_0^{t-1}
12: $w^t \leftarrow \sum_{n \in N_t \cup S} \frac{|D_n|}{|D|} w_n^{t-1}$
13: $\tilde{w}^t \leftarrow$ Adaptive Model Compression
14: **end for**
15: **function** ClientUpdateWithFedProx(c_n, w^{t-1}, μ)
16: $w_n^{t-1} = w^{t-1}$
17: **for** each local training epoch i = 0, 1,...,E **do**
18: **for** each batch b **do**
19: $F_n(w_n^{t-1}) = L_n(w_n^{t-1}) + \frac{u}{2}\left\|w_n^{t-1} - w^{t-1}\right\|^2$
20: $w_n^t = w_n^{t-1} - \eta \nabla F_n(w_n^{t-1}, b)$
21: **end for**
22: **end for**
23: **return** w_n^t
24: **end function**

- Model Aggregation. The server produces an updated global model w^t through the aggregation of the compressed model updates.After aggregation, downward compression is performed on the updated global model by the server, which helps lower the data load for the next round of distribution more effectively(lines 12-14).

The aforementioned process persists until the desired accuracy is attained or a predetermined number of communication rounds have been executed.

4.4. Convergence analysis

In this section, we will present the necessary assumptions and additional notation that we use to prove the convergence guarantee of the DCCF.

Assumption 1 (Conditional L-smoothness)**.** For any w_1 and w_2, there is a constant $L > 0$ for which the loss function satisfies:

$$\left\|\nabla F(w_1) - \nabla F(w_2)\right\| \le L\left\|w_1 - w_2\right\| \tag{10}$$

Assumption 2 (Bounded Variance)**.** Let ξ_n^t is the local training data samples of client n.The stochastic gradient variance is bounded by:

$$\mathrm{E}\left\|\nabla F_n(w_n^t, \xi_n^t) - \nabla F_n(w_n^t)\right\| \le \sigma^2 \tag{11}$$

Assumption 3 (Gradient is bounded)**.** The desired stochastic gradient paradigm is bounded:

$$\mathrm{E}\left\|\nabla F_n(w_n^t, \xi_n^t)\right\| \le G \tag{12}$$

Assumption 4 (Compression operator shrinkage)**.** At round t client n undergoes a compression rate p to obtain the difference between the compression gradient $\tilde{g}_n^t$ and the original gradient g_n^t it can be described as:

$$\|\hat{g}_n^t - g_n^t\| \le \sqrt{(1-p_n^t)}\|g_n^t\| \le \sqrt{(1-p_n^t)}G \tag{13}$$

Assumption 5 (Residual contractility)**.** Residual update satisfies contractility:

$$\|R_n^t\| \le \gamma \cdot \left\|R_n^{t-1}\right\| + (1-p_n^t)G \tag{14}$$

where $0 < \gamma < 1$ is the attenuation coefficient.

$$\|R_n^t\| \le \left\|R_n^{t-1}\right\| + \left\|\Delta W_n^{t-1} - \Delta \widetilde{W}_n^{t-1}\right\| \tag{15}$$

$$\le \left\|R_n^{t-1}\right\| + \sqrt{1-p}G \tag{16}$$

$$\le \gamma \cdot \left\|R_n^{t-1}\right\| + (1-p)G \tag{17}$$

where (15) uses the triangle inequality, (16) utilizes Assumption 4 and (17) controls residual growth via the decay coefficient.

Definition 1. At the t training round, A subset of M clients must be identified such that their aggregated gradient closely approximates the full gradient derived from the original N clients.Therefore,the aggregated gradient difference in the t-th training round is defined to be:

$$v_t = \left\|\sum_{n \in S} \frac{1}{M}\nabla F_n(w_n^t) - \sum_{n=1}^{N} \frac{1}{N}\nabla F_n(w_n^t)\right\| \tag{18}$$

Lemma 1. *For any device k and round t, Under Assumptions 1–5, the single-round variation is bounded by:*

$$\begin{aligned} &F(w^t) - F(w^{t-1}) \\ &\le -\eta\left[\left\|\nabla F(w^t)\right\|^2\right] + \frac{L\eta^2}{2} \cdot (G^2 + \sigma^2) \\ &+ \eta \cdot \left(v_t + \sum_{n \in S} \frac{1}{M}\sqrt{(1-p_n^t)}\left(G + \gamma \cdot \left\|R_n^{t-1}\right\| + (1-p)G\right)\right) \end{aligned} \tag{19}$$

Specifically, in DCCF, since p_n^t is a function of D_n^L and the client set S is also constructed based on D_n^L, their interaction jointly determines the quality of a single round of updates. Proof: See Appendix A.

Theorem 1 (Model Deviation Bound)**.** *Under the above basic assumptions and Lemma 1,We derive the convergence bound by summing loss variations over T rounds:*

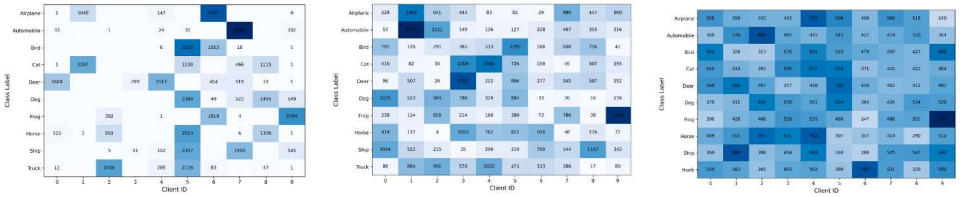
$$\begin{aligned} &E\left(F(w^t) - F(w^{t-1})\right) \\ &\le F(w^0) - F(w^*) + \frac{L\eta^2 T}{2}(G^2 + \sigma^2) \\ &+ \eta \cdot O(T) + \eta \cdot \sum_{t=1}^{T}\sum_{n \in S}\frac{1}{M}\sqrt{(1-p_n^t)} \cdot G \end{aligned} \tag{20}$$

The convergence analysis in this paper further demonstrates that,grounded in selection bias, residual dynamics, round-by-round convergence behavior, and linear growth trends, the cumulative term of DCCF's cooperative optimization mechanism can be bounded by $\mathcal{O}(T)$. By leveraging a unified discrepancy metric, DCCF simultaneously controls both client selection bias and gradient compression error, thereby achieving a tighter upper bound on the objective function. In contrast to traditional sequential strategies (e.g., FedPAQ), DCCF's co-optimized approach not only mitigates error accumulation caused by decoupled decision-making but also dynamically balances the trade-off between bias and compression error through adaptive adjustment of the discrepancy metric. Consequently, the theoretical analysis confirms that DCCF's performance gains arise from its structured cooperative mechanism, rather than from the mere aggregation of independently optimized components.

5. Experiments

5.1. Experiment settings

All experiments were implemented using Python 3.7.2 and the PyTorch framework (Paszke et al., 2019). All experiments were conducted under the same simulated network environment. The Ubuntu 18.04.4 LTS operating system and Tesla T4 GPUs were utilized for model training and evaluation.



(a) α=10 (b) α=1 (c) α=0.1

Fig. 3. Data distribution under CIFAR-10, as α increases the data distribution becomes more similar.

5.1.1. Dataset and model

The performance of DCCF is evaluated on three benchmark datasets: MNIST (Deng, 2012), Fashion-MNIST (Xiao et al., 2017) and CIFAR-10 (Recht et al., 2018). The MNIST dataset comprises images of handwritten digits, with 60,000 training images and 10,000 test images, each being 28 × 28 grayscale. Fashion-MNIST dataset consists of 10 categories such as dresses, jackets and pants, featuring 60,000 training images and 10,000 test images, all in 28 × 28 grayscale.The CIFAR-10 dataset contains 10 categories of color images, with 50,000 training images and 10,000 test images, each sized 32 × 32 and comprising three channels (RGB). We use Logistic Regression model for MNIST, LSTM model for Fashion-MNIST dataset, and CIFAR-10 is trained using modified VGG11 model.

5.1.2. Benchmarks

We have selected five methods as benchmarks for DCCF for comparison. They are as follows:

FedAvg (McMahan et al., 2017) randomly selects clients for aggregation by averaging their model parameters.

FedProx (Li et al., 2020) restricts the divergence between local and global model parameters through the inclusion of proximal regularization.

STC (Sattler et al., 2019) employs sparse quantization techniques to reduce the amount of training iterations and communication budget.

ISFL (Zhu et al., 2024) improves model accuracy and convergence speed through a generalized Importance Sampling Federated Learning framework.

RFCSC (Pan et al., 2025) integrates dynamic client selection with adaptive gradient compression within a reinforcement learning framework. It reduces communication overhead while mitigating the adverse effects of Non-IID data on model convergence.

CSFedProx selects high-quality clients and adds proximal constraint terms the former focuses on better update selection and the latter focuses on update control, both of which synergistically improve model accuracy.

DCCF optimizes client selection and gradient compression strategies by dynamically selecting low-heterogeneity clients and allocating personalized gradient compression ratios.

5.1.3. Training and parameters

The core parameter settings of this experiment adhere to the mainstream research conventions for Non-IID scenarios in federated learning (Sattler et al., 2019; Tang et al., 2022; Zhang et al., 2021). Specifically, the total number of participating clients is set to $K = 20$, each round selects 15 clients to participate in training. Additionally, a parameter $\psi = 5$ is defined, indicating that in each training round, the top 50% of clients with the smallest L2 norm serve as the reference baseline set. At the initial training stage, the initial weight $p_0 = 0.5$ is uniformly assigned to all clients, $p_{\max} = 0.9$ and $p_{\min} = 0.1$. Here, τ_{high} is defined as the maximum weight dispersion within this reference baseline set, while τ_{low} is its minimum value. Compression ratios are dynamically assigned within this interval based on each client's L2 norm, enabling the compression strategy to adapt to the evolving distribution of client models in each round. The residual vector R_0 is defined as the cumulative compensation vector for gradient compression errors, which shares the same dimension as the model gradient. It is initialized as an all-zero vector $R_0 = 0$. Other experimental configurations are set as follows: the local batch size is 40, the learning rate is 0.001, and the SGD optimizer is adopted.

5.2. Analysis of the experimental results

In order to systematically evaluate the robustness of the methods, this study adopts the Dirichlet distribution to construct a differentiated Non-IID data environment (parameter $\alpha \in \{0.1, 1, 10\}$Wang et al., 2024), and the local data distributions of CIFAR-10 as an example are shown in Fig. 3, and validated on multiple benchmark datasets with heterogeneous network architectures.

5.2.1. Client selection strategy

We designed three sets of federated learning pre-experiments on the MNIST dataset to assess the effectiveness of this approach (Fig. 4), incorporating data heterogeneity, with the total number of participating clients fixed at 10. The experimental configurations are as follows: FinalSet-All aggregates only the 10 clients with the lowest L2-norm in each round; FedAvg-Random select 10 clients entirely at random; Hybrid-Select dynamically combines 5 low-dispersion clients with 5 random clients. The experiment ran for 70 rounds with a base learning rate $\eta = 0.01$ and local iteration $E = 20$.

The results indicate that the Hybrid-Select algorithm achieved an average test accuracy of 92.4%, significantly surpassing that of FedAvg-Random (91.5%) and FinalSet-All (91.3%), with respective improvements of 0.9% and 1.1%. Such a result highlights Hybrid-Select's efficacy in dealing with Non-IID data. Although FinalSet-All reduces certain adverse effects, its static client selection might restrict client diversity, thus raising the possibility of getting stuck in local optima.

In contrast, although FedAvg-Random improves participant diversity, it fails to effectively alleviate the bias caused by severe Non-IID data distributions, which leads to oscillatory convergence behavior and marginal gains in accuracy. The Hybrid-Select algorithm, however, ensures stability by selecting low-L2-norm clients while incorporating random clients to enhance diversity, thereby preventing overfitting. Additionally, it optimizes client selection in each iteration, leveraging contributions from random clients to enrich valuable updates. These findings highlight the importance of balancing discrepancy reduction with model diversity, thereby establishing a robust basis for advancing the Hybrid-Select methodology.

5.2.2. Model accuracy

Table 1 compares the model accuracy and communication efficiency of multiple methods under varying Non-IID data heterogeneity levels. Results demonstrate that DCCF significantly reduces communication overhead while maintaining high accuracy, exhibiting stronger robustness as heterogeneity increases. Specifically, as the data distribution becomes increasingly Non-IID (α decreasing from 10 to 0.1), all methods experience accuracy degradation, but DCCF exhibits the smallest decline. For instance, on the CIFAR-10 dataset, DCCF's accuracy decreased from 77.1% to 68.7%, a drop of only 8.4%, whereas FedAvg's accuracy

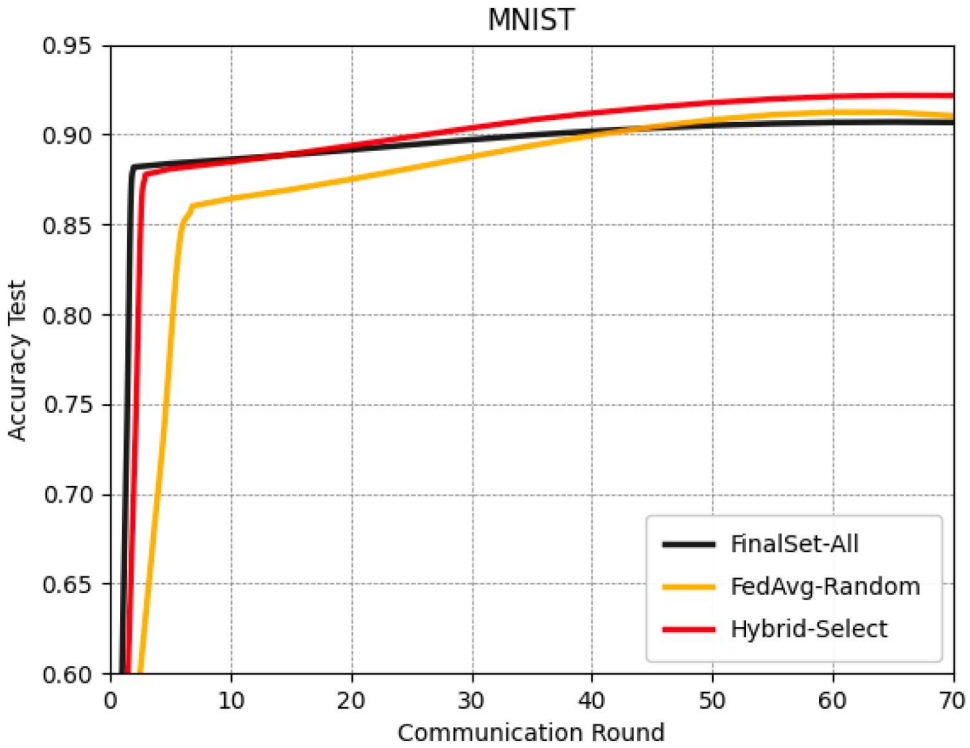
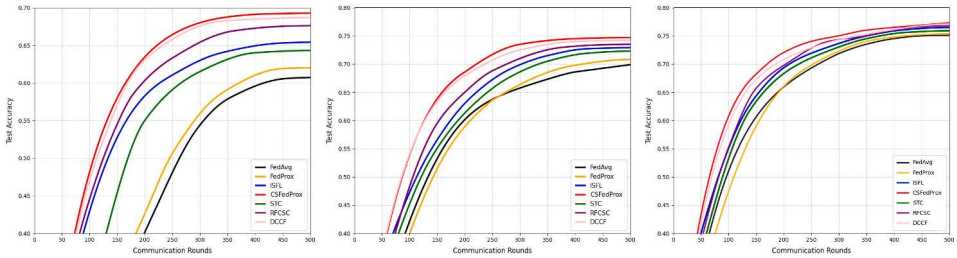


Fig. 4. Accuracy of three sets of client selection schemes on Non-IID data with MNIST dataset.

Table 1
The optimal model accuracy and communication overhead of different methods.

$\alpha = 10$	MNIST		FashionMNIST		CIFAR-10	
	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)
FedAvg	91.7	1.07	87.3	29.3	75.1	709
FedProx	91.6	1.04	87.9	30.7	75.4	786
ISFL	92.3	0.96	88.6	24.3	76.5	662
CSFedProx	**92.5**	0.78	**89.4**	19.7	**77.3**	598
STC	92.2	0.16	88.5	3.8	75.9	130
RFCSC	92.4	0.31	89.1	6.9	76.8	360
DCCF	92.4	**0.08**	89.3	**2.0**	77.1	**69**
$\alpha = 1$	MNIST		FashionMNIST		CIFAR-10	
	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)
FedAvg	90.2	3.81	80.3	120.5	69.8	973
FedProx	90.3	2.63	81.2	114.2	70.7	928
ISFL	91.2	1.98	84.2	57.5	72.9	808
CSFedProx	**91.7**	1.26	**85.9**	33.7	**74.6**	725
STC	91.0	0.23	82.7	7.5	72.3	142
RFCSC	91.4	0.48	85.0	14.6	73.5	450
DCCF	91.6	**0.13**	85.5	**3.9**	74.2	**75**
$\alpha = 0.1$	MNIST		FashionMNIST		CIFAR-10	
	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)	Accuracy(%)	Overhead(MB)
FedAvg	87.8	16.72	75.3	181	60.8	1305
FedProx	88.5	10.17	76.2	173	62.1	1253
ISFL	89.5	8.21	78.6	155	65.2	957
CSFedProx	**90.3**	7.23	**81.2**	106	**69.3**	904
STC	89.4	1.21	77.1	23.3	64.9	192
RFCSC	89.8	3.62	79.9	53.5	67.6	545
DCCF	90.2	**0.65**	80.7	**9.6**	68.7	**103**

fell from 75.1% to 60.8%, representing a 14.3% decline. This demonstrates that DCCF better mitigates performance degradation caused by data heterogeneity. This resilience stems from DCCF's synergistic optimization mechanism combining dynamic client selection and personalized compression. It synchronously filters clients with high consistency using a discrepancy metric while assigning low compression ratios to clients with high discrepancy to preserve critical gradients, thereby maintaining high accuracy even under extreme Non-IID conditions. In terms of communication efficiency, DCCF also demonstrates outstanding performance. At $\alpha = 0.1$, its communication overhead is only 103 MB, significantly lower than FedAvg (1305 MB) and STC (192 MB). This advantage is further highlighted when compared to RFCSC, another



(a) α=0.1 (b) α=1 (c) α=10

Fig. 5. Model convergence process under CIFAR-10 dataset.

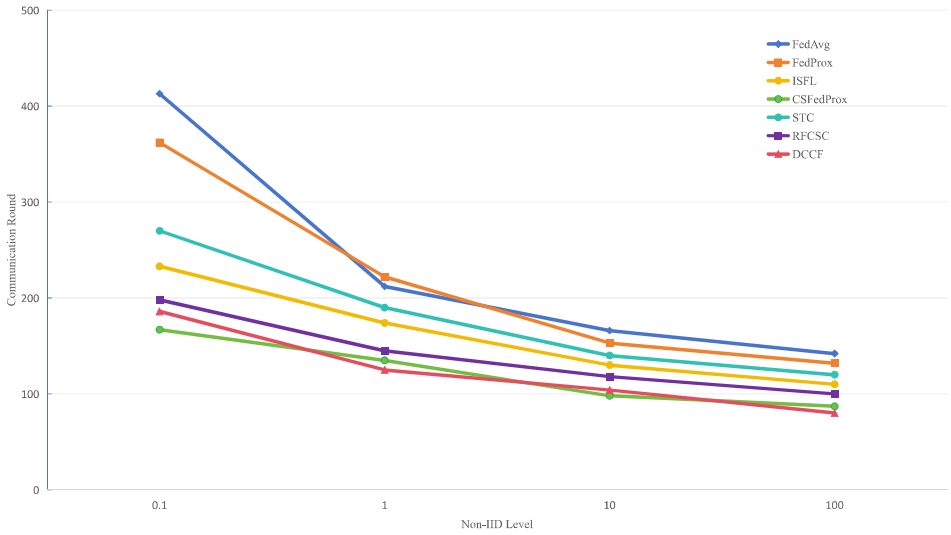


Fig. 6. Communication rounds needed to reach target accuracy under varying heterogeneity levels on CIFAR-10.

method that also pursues joint optimization. DCCF achieves higher accuracy and significantly lower communication overhead than RFCSC. The superiority stems from DCCF's lightweight, discrepancy-aware adaptive personalized compression strategy. Compared to CSFedProx (69.3%, 904 MB), which achieves similar accuracy, DCCF reduces communication overhead by approximately 88% while only sacrificing 0.6 percentage points in accuracy. It is important to note that CSFedProx is designed to maximize model accuracy through refined client selection at the cost of high communication overhead, whereas DCCF is engineered for effectively balancing precision and efficiency.

5.2.3. Convergence

To further evaluate the convergence performance under different Non-IID conditions, a comparative analysis was conducted under CIFAR-10 dataset. As depicted in Fig. 5, on the CIFAR-10 dataset ($\alpha = 0.1$), DCCF converges to stability in approximately 350 rounds, demonstrating faster convergence than RFCSC (370 rounds), ISFL (400 rounds), STC (380 rounds), and FedAvg (480 rounds). This is attributed to the adaptive client selection method based on discrepancy metrics. By dynamically evaluating the Discrepancy between clients and the global model in real-time, it selects the most representative clients for each training round. This effectively mitigates gradient conflicts and directional biases caused by data heterogeneity, enabling stable convergence with fewer communication rounds. Although DCCF exhibits slightly lower final accuracy than CSFedProx under identical conditions (within 1% difference), its convergence process is significantly faster. DCCF's sparsification inevitably introduces a small amount of information loss, while CSFedProx's uncompressed updates preserve all gradient details. However, DCCF effectively compensates for this information loss through its residual accumulation mechanism, ensuring reduced accuracy loss while achieving outstanding communication efficiency. This further demonstrates that DCCF effectively offsets information loss from gradient compression through its residual accumulation compensation mechanism, achieving a favorable balance between convergence speed and model accuracy.

5.2.4. The influence of communication rounds on Non-IID data

To evaluate the framework's robustness to statistical heterogeneity, Fig. 6 quantifies the communication rounds required for 60% precision on CIFAR-10 across Non-IID levels. As data heterogeneity intensifies (α decreasing from 100 to 0.1), the number of communication rounds required for DCCF to achieve target accuracy only increases gradually from approximately 70 to 176 rounds–a growth rate lower than that of all baseline methods and RFCSC, which also adopts a joint optimization approach. This strong robustness stems from the tightly coupled joint optimization mechanism of client selection and gradient compression, which coordinates client selection and personalized compression rate allocation through a unified discrepancy metric at each iteration. This approach maintains high information utility in aggregated updates despite varying heterogeneity, avoiding the convergence efficiency loss caused by critical gradient loss in traditional sequential schemes. Consequently, DCCF achieves optimal communication rounds and stability in Non-IID environments.

5.2.5. Communication efficiency

To systematically assess DCCF's communication performance, we measure how much overhead is needed to achieve a predefined test accuracy across various models and datasets. As shown in Fig. 7, DCCF demonstrated optimal communication efficiency in this evaluation, with significantly lower overhead than all benchmark methods, including

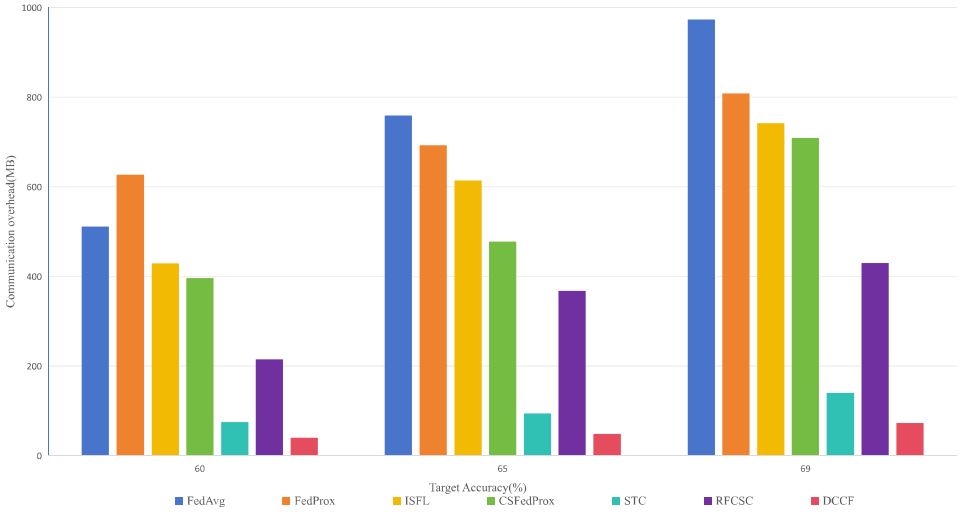


Fig. 7. Communication overhead of various methods for achieving target accuracy on CIFAR-10.

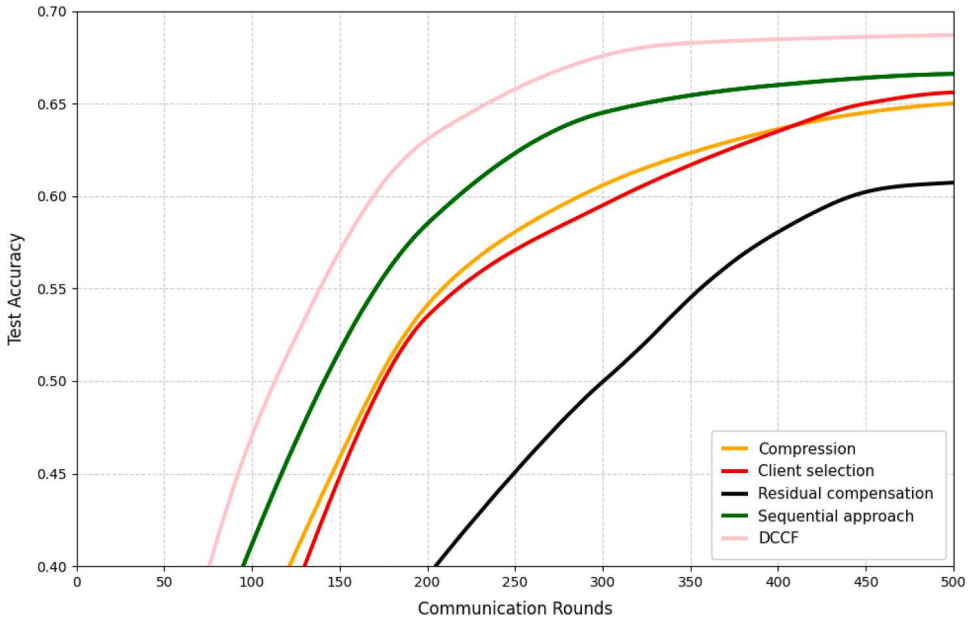


Fig. 8. Ablation study on the impact of DCCF modules on model performance.

STC and RFCSC. This gain stems from its discrepancy-aware personalized compression scheme, which allocates adaptive compression ratios to clients based on their specific discrepancy metrics. The framework reduces per-round communication load synergistically across two key dimensions: the number of participating clients and the bits per gradient element via low-bit quantization, all while preserving critical information. Unlike RFCSC, which follows a sequential selection-compression pipeline, DCCF's unified discrepancy metric enables tighter integration of the two decisions, leading to more efficient resource utilization.

5.3. Ablation analysis

To systematically evaluate the contribution of each core component in DCCF, we conduct an ablation study on the CIFAR-10 dataset.

5.3.1. Necessity of joint optimization

As shown in Fig. 8, on the extremely heterogeneous dataset CIFAR-10 ($\alpha = 0.1$), the standalone gradient compression strategy achieves an accuracy of approximately 65% with slow convergence. Although relying solely on client selection strategies can enhance training stability, the requirement to transmit full model updates results in significant communication overhead. It is noteworthy that the traditional sequential combination strategy (select first, then compress) achieves only 65.3% accuracy, indicating that a simple serial combination of selection and compression fails to resolve the inherent conflict between their objectives under Non-IID data. In contrast, DCCF synchronously coordinates client selection and compression decisions through a unified discrepancy metric. It achieves the same accuracy while reducing communication rounds by 40-50% and lowering communication overhead to 7.9% of the baseline. This performance advantage does not stem from the independent contributions of components such as residual accumulation or adaptive compression, but rather benefits from a collaborative optimization mechanism based on a unified discrepancy metric. This mechanism maximizes communication efficiency while preserving critical gradient information.

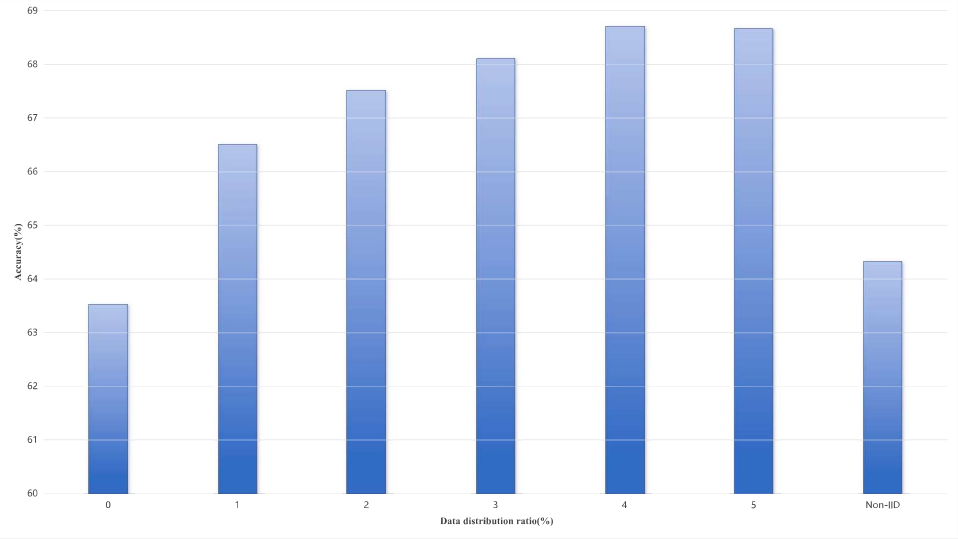


Fig. 9. Sensitivity of DCCF to auxiliary dataset scale and distribution.

5.3.2. Analysis of the impact of auxiliary datasets

The auxiliary dataset in our framework acts as a common reference for estimating the discrepancy between local and global models. To assess the sensitivity of DCCF to the scale and distribution of this data, we conduct an ablation study (Fig. 9). Using a subset of CIFAR-10's training set as an approximately IID public dataset, we vary its proportion from 1% to 5%, while the remaining data is distributed in an extremely Non-IID manner ($\alpha = 0.1$) across 100 clients. Experimental results yield two notable insights into DCCF's performance with respect to auxiliary data. On one hand, DCCF shows low sensitivity to the scale of auxiliary data: even with a mere 1% of auxiliary data, the framework maintains a test accuracy of over 66%, and this metric stabilizes at approximately 68.7% once the auxiliary data proportion reaches 4%. This outcome confirms that a minimal volume of auxiliary data (less than 5%) is adequate to support reliable discrepancy metric computation. On the other hand, DCCF exhibits strong robustness to distributional imperfections in the auxiliary dataset; even when the auxiliary data itself deviates from perfect IID distribution (e.g., with a moderate heterogeneity level of $\alpha = 1$), DCCF still achieves performance that significantly outperforms the baseline setting with no auxiliary data deployed, demonstrating the framework's robustness to distribution defects in auxiliary data. This analysis further validates the feasibility of introducing a small-scale, approximately IID common dataset to evaluate the heterogeneity of client contributions in Non-IID scenarios (Ma et al., 2023; Qi et al., 2023; Zhao et al., 2018).

6. Conclusion

This paper proposes the DCCF framework to address three major challenges posed by non-iid Data: model drift, slow convergence, and high communication overhead. Unlike the traditional sequential paradigm of "select first, compress later," DCCF achieves dynamic coordination between client selection and gradient compression through a unified discrepancy metric. This metric simultaneously evaluates the discrepancy between clients and the global model, driving two coupled decisions: adaptive screening balances model consistency and data diversity to suppress model drift, while personalized compression ratios are assigned to selected clients–high compression for low-discrepancy clients to conserve bandwidth, and low compression for high-discrepancy clients to preserve critical information. Additionally, a residual accumulation compensation mechanism further enhances training stability. We acknowledge several limitations in this work. Scaling our approach to large-scale real-world scenarios remains challenging. Future research will explore scalability solutions for hyper-scale federated learning systems, including but not limited to hierarchical filtering mechanisms and dynamic resource scheduling strategies.

CRediT authorship contribution statement

Xuehua Sun: Conceptualization, Methodology, Software, Writing – Original Draft, Writing - Review & Editing, Investigation.

Chenglong Wu: Software, Validation, Formal analysis, Data Curation, Writing – Review & Editing.

Xianguang Kong: Supervision, Project administration, Resources, Writing – Review & Editing, Funding acquisition.

Guowei Zhang: Methodology, Formal analysis, Validation, Writing – Review & Editing, Investigation.

Runmeng Du: Data Curation, Resources, Software, Writing – Review & Editing.

Wei Sun: Writing – Review & Editing, Supervision, Validation, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of lemma 1

By Assumption 1, the process of approximating the global loss function using Taylor expansion is as follows:

$$\mathrm{E}\left(\mathrm{F}\left(w^{t}\right)-\mathrm{F}\left(w^{t-1}\right)\right) \geq \mathrm{E}\left[\left\langle \nabla \mathrm{F}\left(w^{t}\right), w^{t-1}-w^{t}\right\rangle-\frac{L}{2}\left\|w^{t}-w^{t-1}\right\|^{2}\right] \tag{A.1}$$

The new global model is as follows:

$$\mathrm{W}^{t}-\mathrm{W}^{\mathrm{t}-1}=\frac{\eta}{M} \sum_{n \in S}\left(\nabla \mathrm{F}_{\mathrm{n}}\left(w^{t}\right)+R_{\mathrm{n}}^{\mathrm{t}-1}\right) \tag{A.2}$$

By substituting (A.2) into (A.1) yields:

$$\begin{aligned} \mathrm{F}\left(w^{t}\right)-\mathrm{F}\left(w^{t-1}\right) \leq & -\frac{\eta}{M} \underbrace{\left\langle \nabla \mathrm{F}\left(w^{t}\right), \sum_{n \in S}\left(\nabla \mathrm{F}_{n}\left(w^{t}\right)+R_{\mathrm{n}}^{\mathrm{t}-1}\right)\right\rangle}_{A} \\ & +\frac{L \eta^{2}}{2 M^{2}} \cdot\left\|\sum_{n \in S}\left(\nabla \mathrm{F}_{n}\left(w^{t}\right)+R_{\mathrm{n}}^{\mathrm{t}-1}\right)\right\|^{2} \end{aligned} \tag{A.3}$$

Then, by utilizing Assumptions 3-4, Definition 1 and the Cauchy-Schwarz inequality. A in (A.4) is bounded by:

$$\begin{aligned} A= & \left\langle \nabla \mathrm{F}\left(w^{t}\right), \sum_{n \in S}\left(\nabla \mathrm{F}_{n}\left(w^{t}\right)+R_{\mathrm{n}}^{\mathrm{t}-1}\right)\right\rangle \\ \leq & M\left(\left\|\nabla F\left(w^{t}\right)\right\|+\nu_{t}\right) \\ & +\left\|\nabla F\left(w^{t}\right)\right\| \cdot \sum_{n \in S}\left(1-p_{n}^{t}\right) G \\ & +\left\|\nabla F\left(w^{t}\right)\right\| \cdot\left(\sum_{n \in S} \gamma \cdot E\left[\left\|R_{\mathrm{n}}^{t-2}\right\|\right]+\left(1-p_{n}^{t}\right) G\right) \end{aligned} \tag{A.4}$$

By utilizing Assumption 2 and the triangle inequality:

$$\left\|\sum_{n \in S}\left(\nabla \mathrm{F}_{n}\left(w^{t}\right)+R_{\mathrm{n}}^{t-1}\right)\right\|^{2} \leq M^{2} \cdot\left(G^{2}+\sigma^{2}+\gamma^{2} \cdot\left\|R_{\mathrm{n}}^{t-1}\right\|^{2}+(1-p)^{2} G^{2}\right) \tag{A.5}$$

By substituting (A.4), (A.5) into (A.4), we have

$$\begin{aligned} & E\left[F\left(w^{t}\right)-F\left(w^{t-1}\right)\right] \\ & \leq-\eta E\left[\left\|\nabla F\left(w^{t}\right)\right\|^{2}\right]+\frac{L \eta^{2}}{2} \cdot\left(G^{2}+\sigma^{2}\right) \\ & +\eta \cdot\left(\nu_{t}+\sum_{n \in S} \frac{1}{M} \sqrt{\left(1-p_{n}^{t}\right)}\left(G+\gamma \cdot\left\|R_{\mathrm{n}}^{t-1}\right\|+(1-p) G\right)\right) \end{aligned} \tag{A.6}$$

Appendix B. Proof of theorem 1

For t = 1,2,...T,we have

$$\begin{aligned} & \sum_{t=1}^{T} E\left[F\left(w^{t}\right)-F\left(w^{t-1}\right)\right] \\ & \leq-\eta \sum_{t=1}^{T} E\left[\|\nabla F\left(w^{t}\right)\|^{2}\right]+\frac{L \eta^{2}}{2} \cdot\left(G^{2}+\sigma^{2}\right) \\ & +\eta \cdot \sum_{t=1}^{T}\left(\nu_{t}+\sum_{n \in S} \frac{1}{M} \sqrt{\left(1-p_{n}^{t}\right)}\left(G+\gamma \cdot\left\|R_{\mathrm{n}}^{t-1}\right\|+(1-p) G\right)\right) \end{aligned} \tag{B.1}$$

Applying "telescopic summation" and simplification yields:

$$\begin{aligned} & F\left(w^{T}\right)-F\left(w^{0}\right) \\ & \leq-\eta \sum_{t=1}^{T} E\left[\|\nabla F\left(w^{t}\right)\|^{2}\right]+\frac{L \eta^{2}}{2} \cdot\left(G^{2}+\sigma^{2}\right) \\ & +\eta \cdot \sum_{t=1}^{T}\left(\nu_{t}+\sum_{n \in S} \frac{1}{M} \sqrt{\left(1-p_{n}^{t}\right)}\left(G+\gamma \cdot\left\|R_{\mathrm{n}}^{t-1}\right\|+(1-p) G\right)\right) \end{aligned} \tag{B.2}$$

Based on the convexity assumption, we have

$$E\left[F\left(w^{t}\right)-F\left(w^{*}\right)\right] \leq E\left[\left\langle \nabla F\left(w^{t}\right), w^{t}-w^{*}\right\rangle\right] \tag{B.3}$$

The initial residual $\left\|R_{\mathrm{n}}^{t-1}\right\|$ is bounded, and the compression ratio p satisfies $\sum_{t=1}^{T}\left(1-\mathrm{p}_{n}^{t}\right)=\mathcal{O}(T)$,we have

$$\sum_{t=1}^{T}\left\|R_{\mathrm{n}}^{t-1}\right\| \leq \mathcal{O}(T) \tag{B.4}$$

Combine convexity and upper bound of residuals,we have

$$E\left[\left(F\left(w^{T}\right)-F\left(w^{*}\right)\right)\right]=E\left[F\left(w^{T}\right)-F\left(w^{0}\right)+F\left(w^{0}\right)-F\left(w^{*}\right)\right] \tag{B.5}$$

By substituting (B.4), (B.5) into (B.2), the proof of Theorem 1. is finished

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